

ISIN 325 Portfolio

Colton Schmid

Ferris State University

ISIN 325

Professor Greg Gogolin

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Table of Contents

DB Design Assignment 1	2
DB Design 1 – Install and Populate	4
SQL 1	6
Backup/Recovery	12
SQL 2	19
Database Security.....	22
DB Standards	24
Database Design 2	27
SQL 3	30

ISIN 325

Database security taught me many things about SQL and Oracle and how databases operate and function. My favorite thing from this class has definitely been all of the Microsoft SQL assignments. This is because I found this software very user friendly and enjoyable. Below is all the assignments that I have completed in the class.

DB Design Assignment 1

This is a football database that contains information about both teams and players. There are bridge tables that connect many-to-many relationships and foreign keys to connect one-to-many relationships. The arrows indicate keys connecting to their foreign keys or bridge tables. In the tables, there are the keys and data types, including integers and VarChar.

Standards

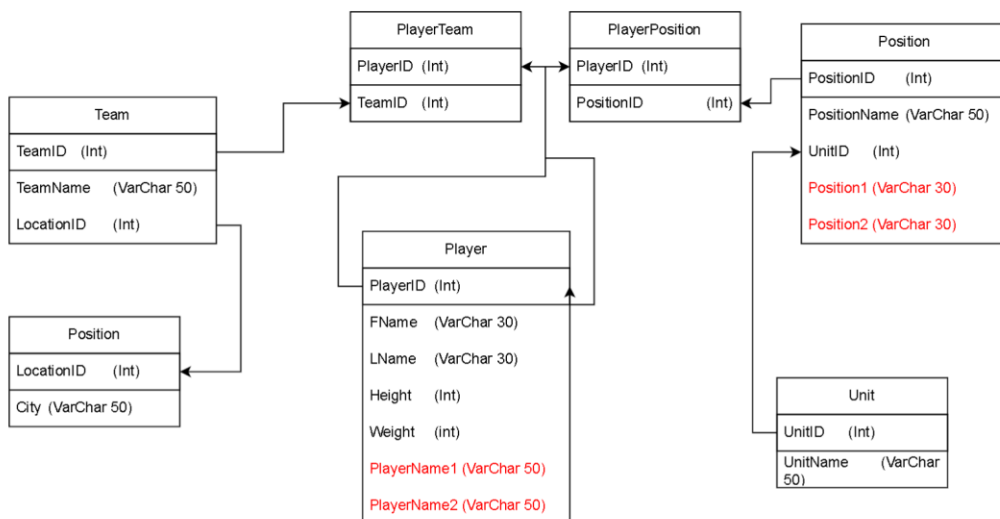
Naming: table names are in lower case and the first character of each word is in UPPER case. There are no spaces between words. This is often referred to as camel case. For example: EmployeeInfo. Column names are in lower case and the first character of each word is in UPPER case. There are no spaces between words. For example: EmployeeId.

Data types: Proper data types are important for good performance. The following guidelines apply to this class: 1) no use of image or complex data types without prior approval of the instructor. 2) data types in table definitions must correspond with data types in SQL routines and stored procedures.

Columns: Columns must not have multiple fields in a single column (i.e. first name and last name should be separate fields - atomic).

Indexes: No more than 3 indexes per table in an OLTP (transactional database). More than 3 indexes per table are allowed in an OLAP (decision support) environment.

Referential Integrity: Referential Integrity should be enforced by the database definition (constraints &/or triggers), not in Queries.



This database follows the relational database theory (normalization). Firstly, this database has no repeating groups; the red-highlighted data in the Player table shows a violation of this rule. We fixed this by having the FName and LName fields. This allows us not only to have atomic data but also to have no repeating groups. Secondly, the red-highlighted data in the Position table shows a violation in repeating data, as we would have to define offense and defense, which would result in repeating data. We solved this by creating the Unit table to define offense from defense. Lastly, everything relies on the key; the arrows show the connection between primary keys and foreign keys. Without these, the tables would not work and would not have nearly enough information.

In this assignment we were tasked with designing and naming the standards of a database. We decided as a class to do a football database.

This assignment was not too bad; however, my lack of direction following cost me most of my issues. I say this because I thought that this assignment was about populating our Football Database. So I spent my time creating the Excel file and attempting to import the data of the football teams. After struggling with many errors, I finally got the Ferris football roster in the database, and I started to question the assignment. I then looked at the assignment and realized that it was on the restaurant database. So I downloaded the scripts, created the database, and populated it.

This assignment was not particularly challenging, as we were given the scripts for the database; however, writing the scripts ourselves would have been a bit more difficult. That being said, I could do it with a different database if I were using this one as a template or a guide. I also think that going through the assignment in class helped a lot with my understanding.

ColtonsLapto...o.Ingredient ✕ **Populate Resta...OP\ccsch (79)** **Resta**

	IngredientID	Name	Description	Unit
▶ 1		Lettuce	Green Lettuce	1 Pound
2		Tomato	Red Tomato	1 Pound
3		Bread	Loaf	1 Pound
4		Pepper	Black Pepper	1 Shaker
5		Filet	Choice	12 oz
6		Rib Eye	Choice	16 oz
7		NY Strip	USDA	12 oz
8		Porterhouse	USDA	32 oz
9		Potatoes	Idaho	Big
10		Carrots	Baby	Bunch
* 11	NULL	NULL	NULL	NULL

100 %
✓ No issues found

Results
Messages

MenuRecCount	
1	16

MenuIngredientRecCount	
1	12

MenuOrderRecCount	
1	9

OrdersRecCount	
1	200

StaffRecCount	
1	25

IngredientRecCount	
1	10

SQL Script:

```

Populate Resta...OP\ccsch (79)
-- Restaurant Tab - OP\ccsch (51)
ColtonsLapto...o.Ingredient
ColtonsLapto...o.Ingredient
ColtonsLapto...o.Ingredient

07 ('Sea Bass with Lemon Butter Sauce', 39.95);
08 ('King Crab Legs', 24.95);
09 ('Lobster Tail 12oz', 59.95);
10 ('Grilled Asparagus with Parmesan', 12.95);
11 ('Garlic Mashed Potatoes', 9.95);
12 ('Salad', 24.95);
13
14
15 INSERT INTO Staff (FName, LName, MName)
16 VALUES
17 ('Michael', 'David', 'Brown'),
18 ('Emily', 'Grace', 'Miller'),
19 ('James', 'Robert', 'Wilson'),
20 ('Sophia', 'Marie', 'Anderson'),
21 ('Daniel', 'Joseph', 'Taylor'),
22 ('Olivia', 'Rose', 'Thomas'),
23 ('William', 'John', 'Martin'),
24 ('Ava', 'Elizabeth', 'Jackson'),
25 ('Alexander', 'Charles', 'Harris'),
26 ('Mia', 'Jane', 'White'),
27 ('Benjamin', 'Paul', 'Thompson'),
28 ('Charlotte', 'Lucas', 'Garcia');
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ColtonsLapto...t - dbo.Menu		ColtonsLapto...bo.Ingredi
MenuID	Name	Price
1	Filet Mignon 8oz	48.95
2	Ribeye 16oz	55.95
3	New York Strip ...	49.95
4	Porterhouse 24oz	65.95
5	Bone-In Cowbo...	72.95
6	Dry-Aged Toma...	115.95
7	Wagyu Striploin...	125.95
8	Surf & Turf (File...	89.95
9	Prime Rib 16oz	54.95
10	Lamb Chops	42.95
11	Sea Bass with L...	39.95
12	King Crab Legs	74.95
13	Lobster Tail 12oz	59.95
14	Grilled Asparag...	12.95
15	Garlic Mashed ...	9.95
16	Salad	24.95
*	NULL	NULL

ColtonsLapto...nulIngredient	Colton
MenuID	IngredientID
1	3
1	4
1	5
2	3
2	6
2	9
3	2
3	7
4	8
16	1
16	2
16	10
* NULL	NULL

ColtonsLapt...bo.MenuOrder		ColtonsLaptop...enuing	
	MenuID	OrderID	Qty
▶	1	1	1
	1	4	1
	1	15	1
	1	16	1
	2	3	1
	3	2	2
	4	2	1
	4	16	1
	5	16	5
*	NULL	NULL	NULL

ColtonsLapto... - dbo.Orders

ColtonsLapto...bo.MenuOrder

OrderID	Location	StaffID	DateTime
1	Austin	173	2025-01-01 06:...
2	Phoenix	9	2025-01-06 02:...
3	San Diego	26	2025-01-03 06:...
4	Dallas	113	2025-01-02 02:...
5	San Antonio	43	2025-01-07 20:...
6	San Diego	99	2025-01-07 12:...
7	San Diego	184	2025-01-02 23:...
8	New York	147	2025-01-05 01:...
9	Los Angeles	151	2025-01-01 13:...
10	Dallas	70	2025-01-06 05:...
11	San Jose	120	2025-01-03 14:...
12	Houston	95	2025-01-06 06:...
13	Austin	14	2025-01-05 15:...
14	Phoenix	65	2025-01-04 19:...
15	San Jose	150	2025-01-02 01:...
16	Austin	136	2025-01-06 07:...
17	Houston	42	2025-01-07 21:...
18	Phoenix	117	2025-01-02 11:...
19	Austin	27	2025-01-07 11:...
20	New York	199	2025-01-04 22:...
21	San Antonio	42	2025-01-01 17:...
22	Chicago	186	2025-01-03 17:...
23	Phoenix	141	2025-01-03 18:...
24	San Diego	42	2025-01-01 12:...
25	Houston	82	2025-01-04 21:...
26	San Antonio	170	2025-01-04 20:...
27	Austin	181	2025-01-05 01:...
28	New York	48	2025-01-04 07:...
29	San Diego	200	2025-01-02 22:...
30	Houston	52	2025-01-01 12:...
31	Los Angeles	7	2025-01-02 23:...
32	Los Angeles	193	2025-01-04 17:...
33	San Jose	46	2025-01-01 13:...
34	Dallas	86	2025-01-02 02:...
35	Chicago	175	2025-01-01 20:...
36	New York	55	2025-01-07 23:...
37	Chicago	176	2025-01-04 03:...
38	Chicago	170	2025-01-07 19:...

ColtonsLaptop...t - dbo.Staff		ColtonsLaptop...t - dbo.Orders		
	StaffID	FName	LName	MName
▶	1	Michael	David	Brown
	2	Emily	Grace	Miller
	3	James	Robert	Wilson
	4	Sophia	Marie	Anderson
	5	Daniel	Joseph	Taylor
	6	Emma	Rose	Thomas
	7	William	John	Martin
	8	Ava	Elizabeth	Jackson
	9	Alexander	Charles	Harris
	10	Mia	Jane	White
	11	Benjamin	Paul	Thompson
	12	Charlotte	Louise	Garcia
	13	Henry	Christopher	Martinez
	14	Amelia	Lee	Clark
	15	Samuel	Edward	Rodriguez
	16	Lily	Claire	Lewis
	17	Lucas	George	Walker
	18	Hannah	Nicole	Hall
	19	Jackson	Henry	Allen
	20	Isabella	Victoria	Young
	21	Matthew	Anthony	King
	22	Ella	Patricia	Wright
	23	Logan	Thomas	Scott
	24	Abigail	Kathleen	Green
	25	David	Andrew	Adams
*	NULL	NULL	NULL	NULL

SQL 1

This Assignment was multiple challenge prompts to do different things in Microsoft SQL. I uploaded both screenshots of the code and output. We used our class Restaurant Database for this assignment.

1) Display the count of the number of rows in the Orders table.

```
ColtonsLaptop\...nt - dbo.Staff  X ColtonsLaptop...t - dbo.Orders
1  Select OrderCount=count(*) from Orders
```

Results	Messages
OrderCount	
1	200

2) Delete a specified record in the Staff table and prove that it is deleted. (before and after)

--First select the record we want to delete to confirm it is there

```
Select * from staff where StaffID= 26
```

Results

Messages

	StaffID	FName	LName	MName
1	26	Colton	Schmid	C

--Deletes StaffID 26 record

```
Delete from staff where StaffID= 26
```

Messages
(1 row affected)
Completion time: 2025-09-25T17:00:23.6983578-04:00

--Confirms StaffID 26 is deleted

```
select * from staff where StaffID= 26
```

	Results		Messages	
	StaffID	FName	LName	MName

3) Two table implied join qualified with a key value.

--Two Table Join

--Combines data from Staff and Orders Table into a single table

```
Select FName, OrderID, DateTime
From Staff, Orders
where Staff.StaffID= Orders.StaffID
```

Results		Messages	
	FName	OrderID	DateTime
1	Alexander	2	2025-01-06 02:53:46.000
2	Amelia	13	2025-01-05 15:16:25.000
3	William	31	2025-01-02 23:35:35.000
4	Samuel	46	2025-01-07 06:08:33.000
5	William	65	2025-01-02 23:45:11.000
6	Lucas	66	2025-01-04 02:58:59.000
7	Michael	68	2025-01-04 15:12:12.000
8	Mia	70	2025-01-07 18:27:52.000
9	Emma	72	2025-01-03 00:15:31.000
10	Daniel	75	2025-01-01 19:11:14.000
11	Ava	93	2025-01-03 22:17:46.000
12	Abigail	104	2025-01-07 17:22:20.000
13	Isabella	114	2025-01-01 20:47:52.000
14	Ava	124	2025-01-01 19:09:19.000
15	Emma	128	2025-01-03 14:28:45.000
16	Lily	144	2025-01-04 06:50:57.000
17	Amelia	145	2025-01-05 23:05:20.000
18	Sophia	159	2025-01-04 02:10:07.000
19	Lucas	167	2025-01-02 22:25:12.000
20	Henry	170	2025-01-04 10:58:17.000
21	William	172	2025-01-07 15:07:39.000

4) Three table implied join qualified with a key value.

```
--Three Table Join
--Combines data from the Ingredient, MenuIngredient, and Menu Tables into a single table
Select I.Name, M.Name, I.Description
From Ingredient I, MenuIngredient MI, Menu M
Where I.IngredientID= MI.IngredientID
and M.MenuID= MI.MenuID
order by I.Name
```

	Name	Name	Description
1	Bread	Filet Mignon 8oz	Loaf
2	Bread	Ribeye 16oz	Loaf
3	Carrots	Salad	Baby
4	Filet	Filet Mignon 8oz	Choice
5	Lettuce	Salad	Green Lettuce
6	NY Strip	New York Strip 12oz	USDA
7	Pepper	Filet Mignon 8oz	Black Pepper
8	Porterhouse	Porterhouse 24oz	USDA
9	Potatoes	Ribeye 16oz	Idaho
10	Rib Eye	Ribeye 16oz	Choice
11	Tomato	New York Strip 12oz	Red Tomato
12	Tomato	Salad	Red Tomato

5) Use twelve SQL functions (such as SUM). This should be twelve queries with at least one function for each query.

```
-- 12 Functions (use w3schools and TSQL functions)
select sum(Price) as MenuTotal from Menu
Select Avg(price) as MenuAvg from Menu
select Min(price) as CheapestItem from Menu
Select Max(Price) as ExpensiveItem From Menu
set rowcount 1
Select Str(Price) as String from Menu
set rowcount 0
Select Rtrim(price) from menu
Select Reverse('tnaruatseR tseB');
Select Replicate ('We are hiring!',5);
Select Nchar(100) as TheUnicode;
Select Lower('THIS MAKES LETTERS LOWERCASE') as LowerFunction
Select Len(Name) From Menu where MenuID=1
Select ASCII(Name) From menu where MenuID=17
Select Concat(FName, LName) from Staff where StaffID=1
```

	MenuTotal
1	961.15
	MenuAvg
1	56.538235
	CheapestItem
1	9.95
	ExpensiveItem
1	125.95
	String
1	49
	(No column name)
1	48.95
2	55.95
3	49.95
4	65.95
5	72.95
6	115.95
7	125.95
8	89.95
	(No column name)
1	Best Restaurant
	(No column name)
1	We are hiring!We are hiring!We are hiring!We are...
	TheUnicode
1	d
	LowerFunction
1	this makes letters lowercase
	(No column name)
1	16
	(No column name)
1	76
	(No column name)
1	MichaelDavid

6) Write an SQL statement that utilizes 'like' in the where clause.

```
--Selects Staff Members who's name starts with a J
Select FName From Staff
where FName like 'J%'
```

	FName
1	James
2	Jackson

7) Demonstrate the use of the rowcount feature to return 4 rows from a table that has more than 4 rows.

```
--Shows the first 4 rows from the Menu table.
set rowcount 4
Select * from Menu
```

	MenuID	Name	Price
1	1	Filet Mignon 8oz	48.95
2	2	Ribeye 16oz	55.95
3	3	New York Strip 12oz	49.95
4	4	Porterhouse 24oz	65.95

Then demonstrate the use of the rowcount feature to reset the number of rows returned to all of the rows in a table.

```
--Set Row Count Back to 0 to display all rows
Set rowcount 0
select * from Menu
```

	MenuID	Name	Price
1	1	Filet Mignon 8oz	48.95
2	2	Ribeye 16oz	55.95
3	3	New York Strip 12oz	49.95
4	4	Porterhouse 24oz	65.95
5	5	Bone-In Cowboy Ribeye 20oz	72.95
6	6	Dry-Aged Tomahawk 32oz	115.95
7	7	Wagyu Striploin 10oz	125.95
8	8	Surf & Turf (Filet & Lobster)	89.95
9	9	Prime Rib 16oz	54.95
10	10	Lamb Chops	42.95
11	11	Sea Bass with Lemon Butter Sauce	39.95
12	12	King Crab Legs	74.95
13	13	Lobster Tail 12oz	59.95
14	14	Grilled Asparagus with Parmesan	12.95
15	15	Garlic Mashed Potatoes	9.95
16	16	Salad	24.95

Demonstrate an alternative to rowcount that has similar functionality.

```
--Alternative to Rowcount
Select top 10 * from Menu
```

	MenuID	Name	Price
1	1	Filet Mignon 8oz	48.95
2	2	Ribeye 16oz	55.95
3	3	New York Strip 12oz	49.95
4	4	Porterhouse 24oz	65.95
5	5	Bone-In Cowboy Ribeye 20oz	72.95
6	6	Dry-Aged Tomahawk 32oz	115.95
7	7	Wagyu Striploin 10oz	125.95
8	8	Surf & Turf (Filet & Lobster)	89.95
9	9	Prime Rib 16oz	54.95
10	10	Lamb Chops	42.95

8) Create a SQL routine to insert values into the Ingredient table.

```
Insert into Ingredient
( Name, Description, Unit)
Values
('Bacon', 'Maple Bacon', '1 Pound')
Select * from Ingredient where Name like 'Bac%'
--do not run twice or will have duplicates
```

	IngredientID	Name	Description	Unit
1	11	Bacon	Maple Bacon	1 Pound

9) Create a SQL routine to update one middle name in the Staff table.

	StaffID	FName	LName	MName
1	1	Michael	David	Brown
2	2	Emily	Grace	Miller
3	3	James	Robert	Wilson

Update Staff

```
set MName = 'Jack' where StaffID = 1
select * from Staff
```

	StaffID	FName	LName	MName
1	1	Michael	David	Jack
2	2	Emily	Grace	Miller
3	3	James	Robert	Wilson
4	4	Sophia	Marie	Anderson

10) Create SQL to truncate and populate a new table with at least 10 rows using data you supply in the insert statement. Table must have at least 3 columns and is not a table in the Restaurant database.

```
--Use this template for create table
--Creates a new table and primary key
CREATE TABLE [dbo].[Equipment](
    [EquipmentID] [int] IDENTITY(1,1) NOT NULL,
    [Name] [nvarchar](50) NOT NULL,
    [Description] [nvarchar](max) NULL,
    [Cost] [decimal](6, 2) NOT NULL,
    CONSTRAINT [PK_Equipment] PRIMARY KEY CLUSTERED
(
    [EquipmentID] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY = OFF, ALLOW_ROW_LOCKS = ON,
) ON [PRIMARY] TEXTIMAGE_ON [PRIMARY]
GO
truncate Table Equipment
```

Messages

Commands completed successfully.

Completion time: 2025-09-26T21:16:41.8038392-04:00

Insert into Equipment

```
( Name, Description, Cost)
Values
('Oven', 'Convection Oven', '499.99'),
('Utensils', 'Forks, Knives, Spoons, etc.', '99.99'),
('Pans', 'Cooking Pans', '124.99'),
('Pots', 'Cooking Pots', '124.99'),
('Spatulas', 'Metal Spatulas', '49.99'),
('Oven Mitts', 'Cooking Mitts', '19.99'),
('Grill', 'Propane Grill', '499.99'),
('Mixing Stand', 'Ingredient Mixing Stand', '274.99'),
('Dishwasher', 'Dishwasher', '499.99'),
('Commercial Fridge', 'Walk in Fridge', '749.99')
```

Messages Client Statistics

(10 rows affected)

Completion time: 2025-09-26T21:34:26.5960713-04:00

11) Create a SQL statement that inserts 2 Orders for a Staff of your choice and 3 Orders for another Staff of your choice.

```
Insert into Orders
( OrderID, Location, StaffID, DateTime)
Values
('201', 'Austin', '181', '2025-01-20 13:20:10.000'),
('202', 'Austin', '181', '2025-01-20 13:30:00.000'),
('203', 'Dallas', '86', '2025-01-20 13:40:20.000'),
('204', 'Dallas', '86', '2025-01-20 13:10:00.000'),
('205', 'Dallas', '86', '2025-01-20 14:00:10.000')
```

Messages Client Statistics

(5 rows affected)

Completion time: 2025-09-26T22:18:38.0851819-04:00

12) Create a query that uses Group By with the Order and Staff table.

```
--Groups the Staff members by their First Name and counts their orders
Select FName, Count(OrderID) as OrderNum
From Staff, Orders
Where Staff.StaffID = Orders.StaffID
Group by FName
```

Results Messages Client Statistics

	FName	OrderNum
1	Abigail	1
2	Alexander	1
3	Amelia	2
4	Ava	2
5	Daniel	1
6	Emma	2
7	Henry	1
8	Isabella	2
9	Lily	1
10	Lucas	2
11	Mia	1
12	Michael	1
13	Samuel	1
14	Sophia	1
15	William	3

13) Create a SQL statement that utilizes the substring command.

```
--Declares the FirstName Variable and sets its type
Declare @FirstName varchar(40)
--Sets what the Variable contains
Select @FirstName = (LName + ', ' + substring(FName,1,1))
From Staff
--Displays the Variable
print @FirstName
```

Messages Client Statistics

Andrew, D

Completion time: 2025-09-26T21:39:49.2552474-04:00

14) Demonstrate the use of distinct, ascending, descending in a query.

```
--Selects only the FName Column from the Staff Table
select distinct FName
From Staff
--Orders the list in Ascending Alphabetical format
Order by FName asc
```

Results Messages Client Statistics

	FName
1	Abigail
2	Alexander
3	Amelia
4	Ava
5	Benjamin
6	Charlotte
7	Daniel
8	David
9	Ella
10	Emily
11	Emma
12	Hannah
13	Henry
14	Isabella
15	Jackson
16	James
17	Lily
18	Logan
19	Lucas
20	Matthew
21	Mia

```
--Selects only the FName Column from the Staff Table
select distinct FName
From Staff
```

```
--Orders the list in Descending Alphabetical format
Order by FName desc
```

	FName
1	William
2	Sophia
3	Samuel
4	Michael
5	Mia
6	Matthew
7	Lucas
8	Logan
9	Lily
10	James
11	Jackson
12	Isabella
13	Henry
14	Hannah
15	Emma
16	Emily
17	Ella
18	David
19	Daniel
20	Charlotte
21	Benjamin

15) Demonstrate @@version

```
--Shows the Version of Microsoft SQL Server running
Select @@version
```

	(No column name)
1	Microsoft SQL Server 2022 (RTM-GDR) (KB5065221...

16) How many Ingredients are in Ingredient table (number, not the list of records).

```
--Counts the number of Ingredients and Displays them as "NumberOfIngredient"
select Count(*) as NumberOfIngredients from Ingredient
```

	NumberOfIngredients
1	12

18) Which Menu Item had the most Ingredients.

```
--Counts how many times a MenuID has an ingredient
select MenuID, Count(IngredientID) as IngredientCT
From MenuIngredient
group by MenuID
--Displays the Menu Items with the most ingredients
Select * From Menu
Where MenuID in (1,2,16);
```

	MenuID	IngredientCT
1	1	3
2	2	3
3	3	2
4	4	1
5	16	3

MenuID	Name	Price
1	Filet Mignon 8oz	48.95
2	Ribeye 16oz	55.95
3	Salad	24.95

19) Which Staff member had the longest Last Name.

```
--Sets the length of the name to NameLen and orders by it descending
Select LName, Len(LName) as NameLen
From staff
order by NameLen desc
```

	LName	NameLen
1	Christopher	11
2	Elizabeth	9
3	Victoria	8
4	Patricia	8
5	Kathleen	8
6	Anthony	7

20) Write a script that creates the data to have a Menu item with more than one Ingredient as well as a script to display that Menu item and it's Ingredients.

```
Insert into Menu
( Name, Price)
Values
('Loaded Baked Potato','14.95')

Insert into MenuIngredient
(MenuID, IngredientID)
Values
('17','9'),
('17','11'),
('17','4'),
('17','10')
```

```

Select M.Name, I.Name
--States what tables to take from
From Ingredient I, MenuIngredient MI, Menu M
Where I.IngredientID= MI.IngredientID
--Sets IDs equal to eachother
and M.MenuID= MI.MenuID
--Shows only MenuID 17
and M.menuID= 17
--Orders by the Ingredient name
order by I.Name

```

	Name	Name
1	Loaded Baked Potato	Bacon
2	Loaded Baked Potato	Carrots
3	Loaded Baked Potato	Pepper
4	Loaded Baked Potato	Potatoes

Backup/Recovery

ISIN 325 Database Security

Backup & Recovery

By Colton Schmid



Introduction

In this assignment we were tasked to create a backup and recovery plan for the database environment. We are the lead DBA of a large corporation that has around 1500 employees and net sales of \$250 million. Our corporation's standard is Microsoft SQL Server, and the HR system is third-party cloud based. This process will be done in five sections: backup schedule, support processes, recovery processes, evaluation, and the turnover procedure. Additionally, I will be going over other related backup and recovery aspects.

Company Databases

Our companies SQL Server Databases include legacy systems, transactional systems, and analytical data warehouses. Our company also has a cloud based server that hosts the human resources system. Below is an overview of the companies database.

Environment	System Types	Location	Importance
In-house SQL Server	Accounting, Forecasting, Planning, Order Entry, Fulfillment, Data Warehouses	On Site	Critical
Cloud Server	Human Resources	Third-Party Vendor	High

This next table shows the database types along with the backup schedules and estimates. The transactional database is backed up in three different ways, same for the legacy database: fully, differentially (only the recent changes since the last fully backup), and the logs. The data warehouse is backed up fully as well, but also in partitions, backing up each partition every 8 hours. The cloud system is a little different, due to it being a third party vendor we must rely on the vendors' managed backups and weekly exports. To ensure the security of these backups they can be encrypted upon being exported or backed up. This will ensure certain aspects of the CIA triad.

Database Type	Backup Type	Scheduled Frequency	Time Estimate	Retention
Transactional	Full	Daily	1-2 hours	7 days
	Differential	Every 6 hours	15-30 minutes	3 days
	Logs	Every 15 minutes	5 minutes	24 hours
Legacy	Full	Daily	1-2 hours	7 days
	Differential	Every 12 hours	30-45 minutes	3 days
	Logs	Hourly	5-10 minutes	1 day
Data Warehouse	Full	Daily	2-4 hours	7 days
	Partition	Every 8 hours	1-2 hours	3 days
Cloud System	Vendor-managed backup	Daily	N/A	30-90 days
	Export	Weekly	30-60 minutes	90 days

There are similarities and differences between backing up in-house databases and cloud databases. One similarity between backing up on premises databases and cloud databases is that daily backups are regular and necessary. One difference between the two is the way the data is backed up. For the cloud database we simply rely on the vendor and then perform weekly

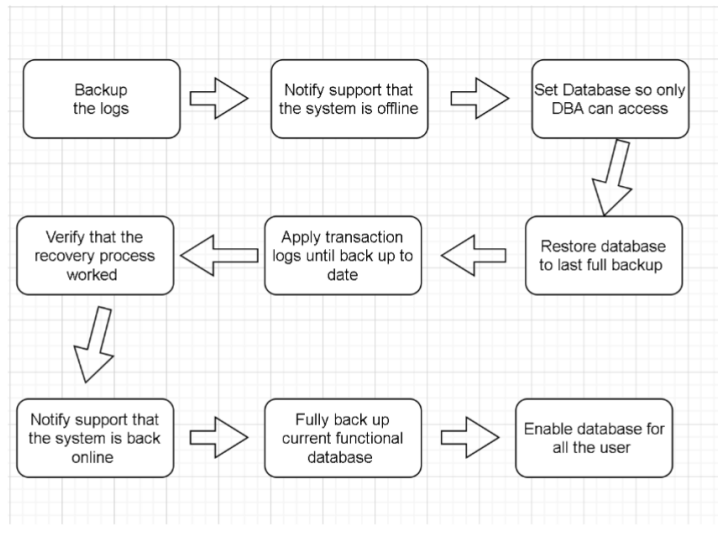
exports, whereas the other database is the data itself. Another difference between these is the cloud system is the only database that has a weekly scheduled backup.

Support Protocols

With as big of a company as we have, it is important to have support protocols in place in case an unfortunate event was to happen to me. Contact information for my position and its responsibilities will be found on site in the correct offices. System prioritization is important in this instance and the replacement should prioritize so that the fluidity of backups continue. However, the new DBA should prioritize the on-site backups as the cloud backups are backed up weekly and less frequently. Staff backups and contingencies are also important, and they cannot be overlooked by the new DBA. These backups need to be stored in a different location in order to maintain all three aspects of the CIA triad.

Recovery Process

The figure below represents the process to recover data in a step-by-step format.



3 | Page

Metrics

- Backup Success Rate? The percentage of scheduled backups vs actual completed backups.
- Time duration of a backup of the database?
- Current database size?
- How long did the recovery process take you?
- How often does the system go down?
- How long was the system down for?
- How much did it cost the corporation?
- Who and or what did it affect?
- How many users were impacted?
- How did you document the procedure.
- How did you complete the recovery?

Turnover/ Change Control

Turnover controls have two different types of controls, simple and complex. Simple turnover controls are when there is a low chance of causing a negative impact on the environment. An example of this would be performing controls at 5 PM every day. Whereas complex controls have a high chance of having a negative impact on the environment if it goes wrong. Below is a change request form.

Change Request Form

[This form is divided into three sections. Section 1 is intended for use by the individual submitting the change request. Section 2 is intended for use by the Project Manager to document/communicate their initial impact analysis of the requested change. Section 3 is intended for use by the Change Control Board (CCB) to document their final decision regarding the requested change.]

1.) SUBMITTER - GENERAL INFORMATION				
CR#	[CR001]			
Type of CR	Enhancement	Defect		
Project/Program/Initiative				
Submitter Name	[John Doe]			
Brief Description of Request	[Enter a detailed description of the change being requested]			
Date Submitted	[mm/dd/yyyy]			
Date Required	[mm/dd/yyyy]			
Priority	Low	Medium	High	Mandatory
Reason for Change	[Enter a detailed description of why the change is being requested]			
Other Artifacts Impacted	[List other artifacts affected by this change]			
Assumptions and Notes	[Document assumptions or comments regarding the requested change]			
Comments	[Enter additional comments]			
Attachments or References	Yes	No		
	Link:			
Approval Signature	[Approval Signature]		Date Signed	[mm/dd/yyyy]

2.) PROJECT MANAGER - INITIAL ANALYSIS			
Hour Impact	[#hrs]	[Enter the hour impact of the requested change]	
Duration Impact	[#dys]	[Enter the duration impact of the requested change]	
Schedule Impact	[WBS]	[Detail the impact this change may have on schedules]	
Cost Impact	[Cost]	[Detail the impact this change may have on cost]	
Comments	[Enter additional comments]		
Recommendations	[Enter recommendations regarding the requested change]		
Technical Approval Signature	[Approval Signature]		Date Signed [mm/dd/yyyy]
Business User Signature	[Approval Signature]		Date Signed [mm/dd/yyyy]
DBA Approval Signature	[Approval Signature]		Date Signed [mm/dd/yyyy]

3.) CHANGE CONTROL BOARD – DECISION				
Decision	Approved	Approved with Conditions	Rejected	More Info
Decision Date	[mm/dd/yyyy]			
Decision Explanation	[Document the CCB's decision]			
Conditions	[Document and conditions imposed by the CCB]			
Approval Signature	[Approval Signature]		Date Signed	[mm/dd/yyyy]

Turnover Procedure

For a simple turnover procedure, the request form would have to be completed and approved by 2pm, the day of the change. Additionally, all necessary files and code must be in place by 2pm, this allows the analysts time to prepare. The business user that can be affected by the change must be included in the approval, as their environment will be affected by the change. With a simple turnover, there is no need to do a full database backup.

For a complex turnover procedure, the request form must be completed and approved by the end of the previous day. Similarly to a simple turnover procedure, the business user that can be affected by the change must approve the request and be informed of the effects of the change on their environment. For a complex turnover procedure, a full backup of the database is required prior to the change, however, a post backup is not always necessary and should be determined by the DBA. The change requester and required resources must be available during the change implementation.

Scripts

A script that determines who is logged into a database is: sp_who

A script that removes/kills a user is: Kill (ID)

A script that backs up the database:

```
BACKUP DATABASE [Restaurant] TO DISK = 'C:\Program Files\Microsoft SQL
Server\MSSQL16.SQLEXPRESS07\MSSQL\Backup\Restaurant.bak' WITH NOFORMAT,
NOINIT, NAME = 'NRestaurant-Full Database Backup', SKIP, NOREWIND, NOUNLOAD,
STATS = 10
```

GO

A script that restores the database:

Use Master

go

ALTER DATABASE Movies

SET RECOVERY FULL;

BACKUP DATABASE Movies

TO DISK = 'C:\Backups\Movies_Backup.bak'

WITH INIT,

7 | Page

FORMAT,

STATS = 10;

BACKUP LOG Movies

TO DISK = 'C:\Backups\Movies_LogBackup.trn'

WITH FORMAT,

NAME = 'Movies Log Backup',

SKIP,

STATS = 10;

RESTORE DATABASE Movies

FROM DISK = 'C:\Backups\Movies_Backup.bak'

WITH NORECOVERY, REPLACE,

STATS = 10;

RESTORE LOG Movies

FROM DISK = 'C:\Backups\Movies_LogBackup.trn'

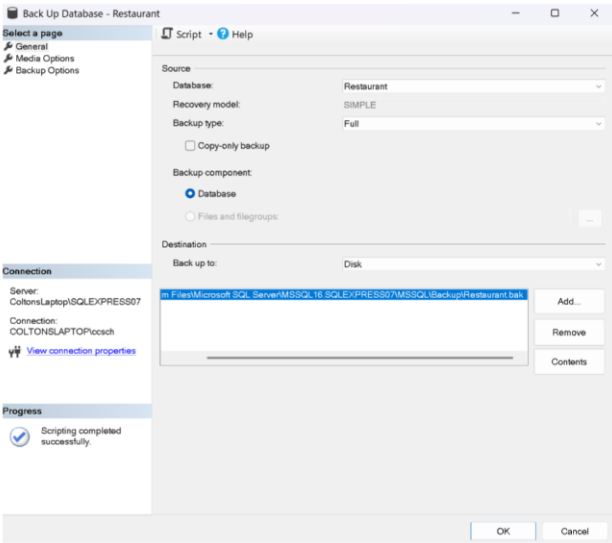
WITH NORECOVERY;

RESTORE DATABASE Movies

WITH RECOVERY;

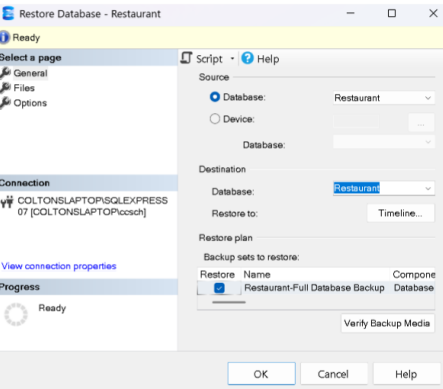
Wizard Backup and Restore

Using Wizard to back up a database:

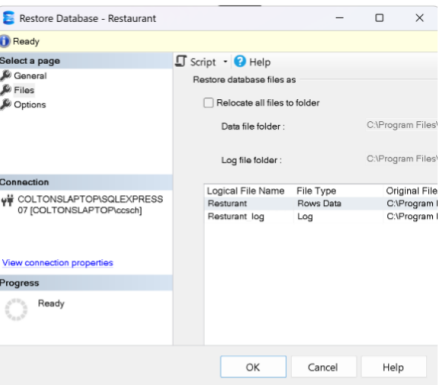


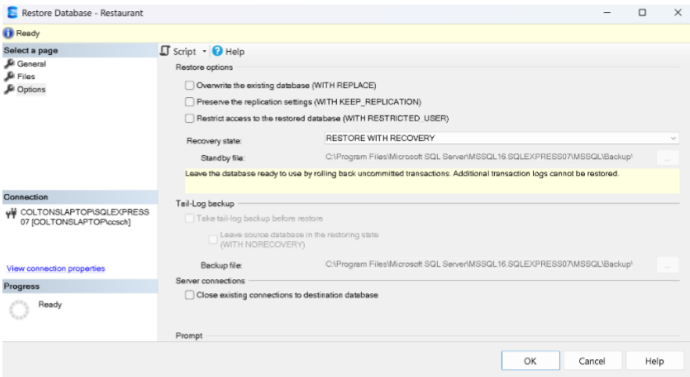
8 | Page

After creating a backup you are now allowed to restore the database back to that backup:



With this page you can also restore the database to a differently named database.

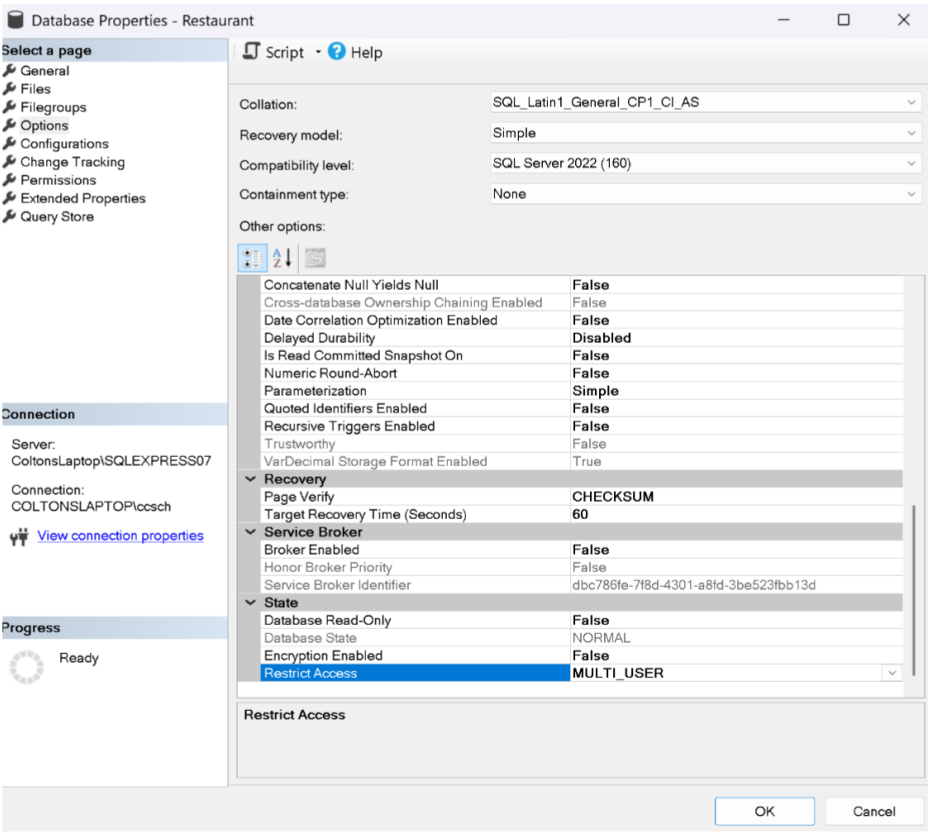




Selecting the option below puts the server in DBO use only. To put it back into multiuser mode

☒ Restrict access to the restored database (WITH RESTRICTED_USER)

You can also go to the databases properties and change access there.



SQL 2

1. Display the count of the number of rows in a table.

```
Select count(*) from Menu
```

Results	Messages
(No column name)	
1	17

2. Delete a specified record in a table and prove that it is deleted.

```
select * from Staff
where StaffID = 25
```

Results	Messages
StaffID FName LName MName	
1 25 David Andrew Adams	

```
Delete from Staff
where StaffID = 25
```

Messages

(1 row affected)

Completion time: 2025-10-15T18:03:42.7601638-04:00

```
select * from Staff
where StaffID = 25
```

Results	Messages
StaffID FName LName MName	

3. Two table join that creates a view.

```
create view vw_GetStaffOrders as
Select staff.StaffID,FName,LName, OrderID
from Staff,Orders
where Staff.StaffID = Orders.StaffID
```

Messages

Commands completed successfully.

Completion time: 2025-10-15T18:53:13.7747676-04:00

4. Three table join qualified with a key value from one view and one other table.

```
Drop view vw_GetStaffOrders
GO
create view vw_GetStaffOrders as
Select staff.StaffID,FName,LName, OrderID, Location
from Staff,Orders
where Staff.StaffID = Orders.StaffID

select Name, GSO.OrderID, Location
From Menu M, MenuOrder MO, vw_GetStaffOrders GSO
where M.MenuID = MO.MenuID
and MO.OrderID = GSO.OrderID
```

Results	Messages
Name OrderID Location	
1 New York Strip 12oz 2 Phoenix	
2 Porterhouse 24oz 2 Phoenix	

ColtonsLapto...etStaffOrders		SQLQuery6.sql - Pyccsch (67)/*		
StaffID	FName	LName	OrderID	Location
1	Michael	David	68	Los Angeles
4	Sophia	Marie	159	Austin
5	Daniel	Joseph	75	New York
6	Emma	Rose	72	San Diego
6	Emma	Rose	128	San Jose
7	William	John	65	Phoenix
7	William	John	31	Los Angeles
7	William	John	172	Phoenix
8	Ava	Elizabeth	124	New York
8	Ava	Elizabeth	93	San Antonio
9	Alexander	Charles	2	Phoenix
10	Mia	Jane	70	New York
13	Henry	Christopher	170	New York
14	Amelia	Lee	145	Chicago
14	Amelia	Lee	13	Austin
15	Samuel	Edward	46	Los Angeles
16	Lily	Claire	144	San Diego
17	Lucas	George	167	Phoenix
17	Lucas	George	66	Los Angeles
20	Isabella	Victoria	114	San Jose
20	Isabella	Victoria	184	San Jose
24	Abigail	Kathleen	104	Dallas
* NULL	NULL	NULL	NULL	NULL

5. Use five SQL functions (such as SUM). This should be five queries with at least one function for each query and not the same functions as you used in previous assignments.

```
Select Lower('THIS MAKES EVERYTHING SMALL')
select ISDATE ('2017-08-25')
Select Soundex('Juice'), SOUNDEX('Juicy')
Select year('2020/08/25') as Year
SELECT CAST(25.65 AS int)
```

Results		Messages
		(No column name)
1	this makes everything small	
		(No column name)
1	1	
		(No column name) (No column name)
1	J200	J200
		Year
1	2020	
		(No column name)
1	25	

6. Create a SQL stored procedure to insert values into a row using variables passed in on execution.

```
Create proc spu_InsertStaff
@FName nvarchar(50),
@LName nvarchar(50)
as
Insert into Staff
(FName, LName)
Values
(@FName, @LName)
Exec spu_InsertStaff 'Jack', 'PF'
```

Messages	
Commands completed successfully.	
Completion time: 2025-10-15T19:11:22.7639032-04:00	

7. Write the SQL to create a table, key, and index. This must not be generated SQL. Example after #12.

```
CREATE TABLE Customers
(CustomerID int NOT NULL,
FName nvarchar(20) NOT NULL,
LName nvarchar(30) NOT NULL,
MName nvarchar(20) NULL,
CONSTRAINT PK_Actor5 PRIMARY KEY CLUSTERED
(CustomerID ASC)
ON [PRIMARY]) -- key
ON [PRIMARY] -- index
```

Messages	
Commands completed successfully.	
Completion time: 2025-10-15T21:47:04.0254558-04:00	

8. Create a query that uses Group By

```
Select FName, Count(OrderID)
from Staff, Orders
where staff.StaffID = Orders.StaffID
group by FName
```

Results		Messages
	FName	(No column name)
1	Abigail	1
2	Alexander	1
3	Amelia	2
4	Ava	2
5	Daniel	1
6	Emma	2
7	Henry	1
8	Isabella	2
9	Lily	1
10	Lucas	2
11	Mia	1
12	Michael	1
13	Samuel	1
14	Sophia	1
15	William	3

9. Demonstrate the use of distinct, ascending, descending in a query. The result set must have at least 7 rows.

```
select Distinct FName
from Staff
Order by FName asc
```

Results		Messages
	FName	
1	Abigail	
2	Alexander	
3	Amelia	
4	Ava	
5	Benjamin	
6	Charlotte	
7	Daniel	
8	Ella	
9	Emily	
10	Emma	
11	Hannah	
12	Henry	
13	Isabella	
14	Jackson	
15	James	
16	Lily	

```
select FName
from Staff
Order by FName desc
```

	FName
1	William
2	Sophia
3	Samuel
4	Michael
5	Mia
6	Matthew
7	Lucas
8	Logan
9	Lily
10	James
11	Jackson
12	Isabella
13	Henry
14	Hannah
15	Emma
16	Emily

10. Create a query that demonstrates an inner join (includes join in statement)

```
Select FName, OrderID, Location
from Staff
      Join Orders
on Staff.StaffID = Orders.StaffID
order by FName
```

	FName	OrderID	Location
1	Abigail	104	Dallas
2	Alexander	2	Phoenix
3	Amelia	13	Austin
4	Amelia	145	Chicago
5	Ava	124	New York
6	Ava	93	San Antonio
7	Daniel	75	New York
8	Emma	128	San Jose
9	Emma	72	San Diego
10	Henry	170	New York
11	Isabella	184	San Jose
12	Isabella	114	San Jose
13	Lily	144	San Diego
14	Lucas	66	Los Angeles
15	Lucas	167	Phoenix
16	Mia	70	New York

11. Create a query that demonstrates a left outer join (includes join in statement)

```
Select FName, OrderID, Location
from Staff
      Left Join Orders
on Staff.StaffID = Orders.StaffID
order by FName
```

	FName	OrderID	Location
1	Abigail	104	Dallas
2	Alexander	2	Phoenix
3	Amelia	13	Austin
4	Amelia	145	Chicago
5	Ava	124	New York
6	Ava	93	San Antonio
7	Benjamin	NULL	NULL
8	Charlotte	NULL	NULL
9	Daniel	75	New York
10	Ella	NULL	NULL
11	Emily	NULL	NULL
12	Emma	128	San Jose
13	Emma	72	San Diego
14	Hannah	NULL	NULL
15	Henry	170	New York
16	Isabella	184	San Jose

12. Create a query that demonstrates a right outer join (includes join in statement)

```
Select FName, OrderID, Location
from Staff
      Right Join Orders
on Staff.StaffID = Orders.StaffID
order by FName
```

	FName	OrderID	Location
1	NULL	14	Phoenix
2	NULL	15	San Jose
3	NULL	16	Austin
4	NULL	17	Houston
5	NULL	18	Phoenix
6	NULL	19	Austin
7	NULL	20	New York
8	NULL	21	San Antonio
9	NULL	22	Chicago
10	NULL	23	Phoenix
11	NULL	24	San Diego
12	NULL	25	Houston
13	NULL	26	San Antonio
14	NULL	27	Austin
15	NULL	28	New York
16	NULL	29	San Diego

Database Security

ISIN 325 Database Security

Database Security

By Colton Schmid



Introduction

In this assignment we were tasked to create a database security policy. We are the lead DBA of a large corporation that has around 1500 employees and net sales of \$250 million. Our corporation's standard is Microsoft SQL Server, and the HR system is third-party cloud based. This assignment will go over the defense in depth of the system, access controls, data access policies, and the auditing and logging process.

Defense in Depth

To defend our databases, we will have three security policies in place to ensure that the CIA triad is fully in place. The first layer is physical security; with this we will have locked rooms for our servers and key cards to access those rooms. Additionally, we will have the physical site of the corporation locked down and only authorized individuals will be able to enter. This ensures that not everyone can just walk into the building or server rooms unauthorized. The second layer that we will implement is accounts and passwords; this ensures

that the account that is doing work on the database is the person qualified and permitted to work on it. Having a secure password system helps eliminate potential breaches and consequential errors. The last layer of security is enabling all the logs, on both of the previous layers specifically. Additionally, we would want to keep the logs for a few weeks, or months, if we have the space for it. This ensures that if there is a breach or attack, we can pinpoint it and figure out the solution faster. These layers are all built on each other and with each layer working properly it can prevent many issues and breaches from happening in the future.

Access Control

With as big of a company as we have, it is important to have access controls in place in order to mitigate errors or unauthorized access to the database. With this our company will use windows security, this lets the users use their windows password for everything. A downside is that if the password gets compromised, the attacker will have access to all of their accounts. With this in place, however, the company will have a password policy that forces the password to be a certain length or longer, as well as contain special characters. In addition to passwords, each user must set up a two-factor authentication method to their accounts. This helps reduce the number of breaches immensely. There will also be a password expiration date, so the user must choose a new password every 90 days. Each account will have a set database to ensure that they are only in the database they are supposed to be in. Additionally, each account will have server roles and database roles. The server roles will dictate what security privileges the account has in the whole system, while the database roles indicate the privileges within the certain database. The database owners would dictate the database roles given out within their database. If a user thinks that they should be able to access a certain database or have a certain privilege, they could make a access request. An access request is a form that the user would fill out and send to their DBO or DBA if necessary. To eliminate creeping authority each users privileges will be job specific and subject to change depending on the users current position.

Data Access

When it comes to data access, users will only be able to access stored procedures and views when required to. A stored procedure is a SQL function that executes a specific task when used. These procedures increase efficiency and security of the database. Access to certain views allows the users to view and sometimes interact with certain tables, however it is not the direct table that they are working on so that increases the security by not allowing them to tamper with the actual data. Our company's policy on embedded SQL is that it is bad and unsecure, so we will not use it and resort to procedures and views instead. When it comes to commercial services, they will be on different isolated servers to ensure that if anything goes wrong with that service it is contained.

Auditing/Logging

When it comes to auditing and logging, we will have logging turned on to everything. Additionally, I would set it up to keep the logs for multiple weeks, if not months, if the system can handle it. I would have it alert and send out daily or half daily reports for failed login attempts, intrusions, unauthorized access attempts, or any other suspicious activity on the network.

DB Standards

ISIN 325 Database Security

Database Standards

By Colton Schmid



Project Overview

In this assignment we were tasked to create a Data Dictionary and a Database Standard. Our organization is a large corporation that has around 1500 employees and net sales of \$250 million. This assignment will go through the database design diagram, data dictionary, and the database standards.

Database Design Diagram



Data Dictionary

With this database there are eight tables, two of which are bridge tables: Menu, Staff, Customers, Equipment, Ingredients, Orders, MenuOrder, MenuIngredient. Menu contains MenuID which is an integer and is the primary key for this table, as well as the foreign key for another table. Menu also contains Name which is a NVarchar datatype and has an index of 50. The price column is a decimal with an index of 6,2. Neither of these are nullable nor keys. Staff contains StaffID which is identical to the MenuID. It also contains FName which is a NVarchar datatype and has an index of 50. Additionally, LName is also a NVarchar datatype with an index of 50. Neither of these are nullable nor keys. MName is nullable, a NVarchar datatype, has an index of 50 and is not a key. The Customers table is identical to the Staff, just set up for customers. Equipment table has EquipmentID which is set up the same way as the other IDs. The Name is a NVarchar datatype and has an index of 50. Description is a NVarchar datatype with the max index. Neither of these are keys but Description is nullable while the Name is not. Cost is set up the same way as Price for the Menu table. The Ingredients table contains Ingredient ID which is also set up the same way as the other IDs. The Name and Description columns work the same as the Equipment table. The Unit column is a NChar with the index of 10. This is not a key, but is it nullable. The Orders table contains OrderID which is also set up like the other IDs. It also contains Location which is a NVarchar with an index of 50. StaffID is a foreign key to the Staff table and is set up the exact same way as StaffID in Staff. DateTime is a datetime datatype and none of these columns are nullable. MenuOrder and MenuIngredient are bridge tables and contain the exact same data as the other tables they connect to. Below is a data dictionary from the database:

	TABLE_NAME	COLUMN_NAME	IS_NULLABLE	DATA_TYPE	CHARACTER_MAXIMUM_LENGTH	CHARACTER_SET_NAME
1	Ingredient	IngredientID	NO	int	NULL	NULL
2	Ingredient	Name	NO	nvarchar	50	UNICODE
3	Ingredient	Description	YES	nvarchar	-1	UNICODE
4	Ingredient	Unit	YES	nchar	10	UNICODE
5	Menu	MenuID	NO	int	NULL	NULL
6	Menu	Name	NO	nvarchar	50	UNICODE
7	Menu	Price	NO	decimal	NULL	NULL
8	MenuIngredient	MenuID	NO	int	NULL	NULL
9	MenuIngredient	IngredientID	NO	int	NULL	NULL
10	MenuOrder	MenuID	NO	int	NULL	NULL
11	MenuOrder	OrderID	NO	int	NULL	NULL
12	MenuOrder	Qty	NO	int	NULL	NULL
13	Orders	OrderID	NO	int	NULL	NULL
14	Orders	Location	NO	nvarchar	50	UNICODE
15	Orders	StaffID	NO	int	NULL	NULL
16	Orders	DateTime	NO	datetime	NULL	NULL
17	Staff	StaffID	NO	int	NULL	NULL
18	Staff	FName	NO	nvarchar	50	UNICODE
19	Staff	LName	NO	nvarchar	50	UNICODE
20	Staff	MName	YES	nvarchar	50	UNICODE
21	sysdiagrams	name	NO	nvarchar	128	UNICODE
22	sysdiagrams	principal_id	NO	int	NULL	NULL
23	sysdiagrams	diagram_id	NO	int	NULL	NULL
24	sysdiagrams	version	YES	int	NULL	NULL
25	sysdiagrams	definition	YES	varbinary	-1	NULL
26	Equipment	EquipmentID	NO	int	NULL	NULL
27	Equipment	Name	NO	nvarchar	50	UNICODE
28	Equipment	Description	YES	nvarchar	-1	UNICODE
29	Equipment	Cost	NO	decimal	NULL	NULL

Database Standards

When it comes to our corporations' standards, we lay them out plain and clear so that all of our operations and activities flow smoothly. Our vendor standards are that every employee uses Microsoft as a vendor. For our design standards we use pre-normal forms and atomic data. For our backup and recovery standards we will back up the databases to an offsite location, as well as have a recovery plan in place so that when we need to make a recovery it will go smoothly. Our naming standards for this database are that everything must be Camel Case format. Our datatype standards include VarChars, decimals, integers, NVarchar, NChar, and datetime. Other datatypes will have to be authorized by a DBA. Our platform standards are that everything is done on the Microsoft SQL Server. The security standards are that employees use stored procedures and views, as these run faster than embedded SQL and more secure. Additionally, our company operates on least privilege. With software standards the company will rely on windows to carry out the operations. Turnover/ Change control standards state how if a turnover or change control is requested that the request must be approved by 3pm the previous day, if not sooner depending on the change.

Summary

While completing this project and reading all of the standards, it is apparent to me that describing these processes in detail will save a company a lot of time when dealing with particular situations. Additionally, distributing this information to new employees will make the transition into the position easier, as they will know what to expect and how they are expected to perform. Ensuring that the organization is on the same page can be crucial to operations as a corporation.

Database Design 2

ISIN 325 Database Security

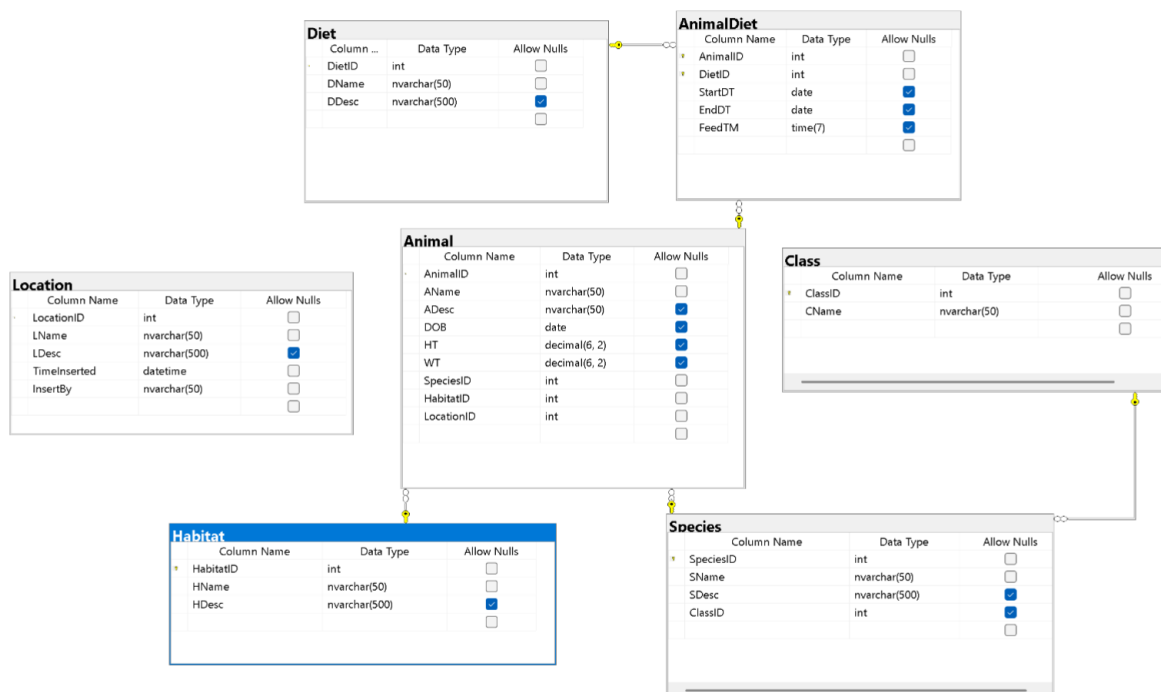
Database Design 2

By Colton Schmid



Database Diagram

In this assignment we were tasked to create a Database that is intended for Animals, with more tables it can be suitable for a zoo database as well. We created seven tables to hold our data: Animal, Location, Class, Diet, Species, Habitat, and AnimalDiet. There are bridge tables and foreign keys to connect the data to the appropriate tables. We populated the tables using SQL scripts and created a functioning database. Below is the Database Design Diagram.



Database Standards

One standard that our database has is the Naming Standard. This standard is that everything is named using CamelCase. For example, the table that contains data pertaining to the animal's diet is named AnimalDiet. Another standard that our database uses is that everything must be created, modified and viewed through Microsoft SQL. Both the database design and database data were created using SQL. One of the last standards that we have is the Datatype standard. With this, specific types of data are required for certain columns. For example, all of our IDs are classified as an integer.

Populated

All of our tables in the database are fully populated and below is the output of each table:

Animal:

	AnimalID	AName	ADesc	DOB	HT	WT	SpeciesID	HabitatID	LocationID
1	1	Simba	Male African Lion	2018-05-12	1.20	190.50	10	1	1
2	2	Nala	Female African Lion	2019-07-08	1.10	165.00	10	1	1
3	3	Dumbo	African Elephant calf	2021-04-15	2.30	450.00	1	1	2
4	4	Ellie	Adult African Elephant	2010-03-09	3.20	5200.00	1	1	2
5	5	Stripe	Male Bengal Tiger	2016-11-22	1.00	220.00	3	2	3
6	6	Tara	Female Bengal Tiger	2018-06-30	0.95	200.00	3	2	3
7	7	Splash	Bottlenose Dolphin	2020-02-10	2.70	180.00	14	4	4
8	8	Fluke	Young Bottlenose Dolphin	2023-06-25	1.90	95.00	14	4	4
9	9	Rocky	Brown Bear	2015-09-12	1.30	300.00	21	8	5
10	10	Bella	Female Brown Bear	2017-05-18	1.10	250.00	21	8	5
11	11	Zara	Female Zebra	2017-11-02	1.40	350.00	22	7	2
12	12	Marty	Male Zebra	2018-03-14	1.50	380.00	22	7	3
13	13	Miko	Chimpanzee	2014-08-10	0.90	45.00	19	6	4
14	14	Bobo	Male Chimpanzee	2016-02-25	0.95	50.00	19	6	5
15	15	Luna	Female Polar Bear	2013-12-18	1.80	450.00	23	5	1
16	16	Blizzard	Male Polar Bear	2012-10-03	1.90	500.00	23	5	2
17	17	Echo	Male Gray Wolf	2015-07-25	0.95	55.00	8	7	1
18	18	Winston	Alpha Gray Wolf	2014-03-19	1.00	60.00	8	7	2
19	19	Marley	Male Koala	2019-04-12	0.60	14.00	6	8	3
20	20	Coco	Female Koala	2020-06-15	0.58	13.50	6	8	4
21	21	Killer	Orca Whale	2011-05-01	6.00	5400.00	13	4	5
22	22	Shadow	Female Orca Whale	2012-03-10	5.80	5100.00	13	4	2
23	23	Penny	Penguin	2020-11-22	0.75	5.50	12	5	1
24	24	Tux	Male Emperor Penguin	2019-02-18	0.80	6.00	12	5	3
25	25	Nova	Giraffe	2017-01-09	4.00	800.00	20	7	2
26	26	Fiona	Female Giraffe	2018-04-25	3.90	750.00	20	7	1
27	27	Ash	Male Red Kangaroo	2020-08-30	2.20	80.00	7	8	5
28	28	Ruby	Female Red Kangaroo	2021-01-15	2.00	75.00	7	8	3
29	29	Leo	Lioness	2016-06-30	1.00	150.00	10	1	2
30	30	Tori	Female Bengal Tiger	2015-12-09	1.10	190.00	3	2	1
31	31	Cleo	Lioness	2018-03-04	1.05	160.00	10	1	3
32	32	Toby	Male American Alligator	2014-07-07	4.00	450.00	15	2	5
33	33	Sally	Female American Alligator	2016-10-17	3.80	400.00	15	2	4
34	34	Theo	Male Cheetah	2017-11-30	1.20	65.00	5	7	1
35	35	Luna	Female Cheetah	2018-02-14	1.10	60.00	5	7	3
36	36	Rex	Male Giraffe	2019-04-03	4.10	850.00	20	7	2
37	37	Sasha	Female Giraffe	2017-06-23	3.90	800.00	20	7	4
38	38	Ace	African Elephant Bull	2010-02-25	3.30	5200.00	1	1	5
39	39	Olive	Female African Elephant	2013-09-09	3.00	4500.00	1	1	2
40	40	Spide	Male Zebra	2015-12-30	1.55	400.00	22	7	5
41	41	Koda	Male Grizzly Bear	2016-05-14	1.50	350.00	21	8	1
42	42	Raven	Female Grizzly Bear	2017-03-20	1.40	320.00	21	8	2
43	43	Toby	Male Polar Bear	2014-06-05	1.85	480.00	23	5	3
44	44	Nina	Female Polar Bear	2015-02-18	1.80	450.00	23	5	4
45	45	Blake	Male Giraffe	2016-01-11	4.00	700.00	20	7	2

Species:

	SpeciesID	SName	SDesc	ClassID
1	1	African Elephant	Largest land animal, native to Africa	1
2	3	Bengal Tiger	Large striped cat native to India and surrounding r...	1
3	5	Cheetah	Fastest land animal, spotted cat native to Africa	1
4	6	Koala	Tree-dwelling marsupial native to Australia	1
5	7	Red Kangaroo	Largest marsupial, native to Australia	1
6	8	Gray Wolf	Large pack-hunting canine found in North America ...	1
7	10	African Lion	Large social cat native to Africa, apex predator	1
8	12	Emperor Penguin	Largest penguin species, native to Antarctica	3
9	13	Orca Whale	Largest dolphin species, apex ocean predator	1
10	14	Bottlenose Dolphin	Intelligent marine mammal found worldwide	1
11	15	American Alligator	Large reptile native to southeastern United States	2
12	19	Chimpanzee	Great ape native to central Africa, highly intelligent	1
13	20	Giraffe	Tallest land animal with long neck, native to Africa	1
14	21	Brown Bear	Large bear species found across North America a...	1
15	22	Zebra	Striped equine native to Africa	1
16	23	Polar Bear	Large bear adapted to Arctic environment	1

Class:

	ClassID	CName
1	1	Mammalia
2	2	Reptilia
3	3	Aves

Habitat:

	HabitatID	HName	HDesc
1	1	Savanna	Open grassland with scattered trees, warm climate
2	2	Tropical Rainforest	Dense forest with high rainfall and humidity
3	3	Temperate Forest	Forest with distinct seasons and deciduous trees
4	4	Aquatic	Water-based habitat for marine and freshwater sp...
5	5	Arctic/Polar	Cold climate with ice and snow
6	6	Jungle	Dense tropical forest with diverse vegetation
7	7	Grassland	Open plains with grasses and few trees
8	8	Woodland	Mixed habitat with trees and open areas

Diet:

	DietID	DName	DDesc
1	1	Carnivore Diet	Composed of raw meats such as beef, chicken, and fish, supplemented with bone meal and vitamins to maintain strength and energy in predatory animals like lions and tigers.
2	2	Herbivore Diet	A high-fiber blend of grasses, hay, and leafy greens with added minerals to promote digestion and health in grazers like zebras, giraffes, and elephants.
3	3	Omnivore Diet	A varied combination of meats, grains, vegetables, and fruits designed to meet the nutritional needs of omnivores such as bears, pigs, and primates.
4	4	Piscivore Diet	Contains fresh or frozen fish and shellfish, offering high protein for penguins, seals, and other fish-eating species.
5	5	Insectivore Diet	Includes live crickets, mealworms, and beetles enriched with calcium and vitamins, suitable for anteaters, frogs, and small reptiles.
6	6	Frugivore Diet	A mix of tropical fruits like bananas, mangoes, and papayas, ideal for fruit-eating species such as bats, monkeys, and toucans.
7	7	Granivore Diet	Formulated from a mix of seeds, nuts, and grains providing essential fats and proteins for parrots, finches, and rodents.
8	8	Nectarivore Diet	A sweetened nectar formula with fruit puree and amino acids designed for hummingbirds and lorikeets.
9	9	Folivore Diet	Primarily composed of eucalyptus leaves, bamboo, and other fibrous greens suited for koalas, pandas, and sloths.
10	10	Detritivore Diet	Includes decaying plant matter, fungi, and compost to sustain decomposers like millipedes and certain beetles.
11	11	Piscivorous Bird Diet	A seafood-based diet featuring small fish and crustaceans to support seabirds such as pelicans and herons.
12	12	Reptile Carnivore Diet	Raw meat and whole prey items such as mice and chicks formulated for snakes, lizards, and crocodiles.
13	13	Amphibian Diet	A moist, insect-heavy blend including worms and flies designed for frogs, salamanders, and toads.
14	14	Marine Mammal Diet	Consists of fresh fish, squid, and vitamin supplements tailored for seals, sea lions, and dolphins.
15	15	Avian Herbivore Diet	A mix of leafy greens, fruits, and seeds suited for parrots, macaws, and other plant-eating birds.
16	16	Primate Diet	Balanced portions of fruits, vegetables, nuts, and proteins for monkeys, lemurs, and apes to support energy and cognition.
17	17	Rodent Diet	A combination of grains, seeds, and fresh vegetables providing fiber and energy for rats, guinea pigs, and squirrels.
18	18	Ungulate Diet	Specialized hay and mineral-enriched grain mix for hoofed animals like antelopes, deer, and bison.
19	19	Arthropod Diet	Comprised of decaying leaves, fruit scraps, and calcium dusted insects for tarantulas, scorpions, and beetles.
20	20	Piscivore Reptile Diet	High-protein fish and crustacean mix developed for aquatic reptiles such as turtles and caimans.
21	21	Seed and Fruit Mix Diet	Combination of seeds, nuts, and fruit pieces designed to provide variety for avian birds and small mammals.
22	22	Crustacean Diet	A diet of algae, fish flakes, and plant matter suitable for lobsters, crabs, and other crustaceans in aquatic exhibits.
23	23	Nocturnal Mammal Diet	A mix of fruits, insects, and small prey designed for nocturnal animals like kinkajous and civets.
24	24	Desert Herbivore Diet	Dried grasses, succulents, and fiber-rich feed created for camels, gazelles, and desert-dwelling tortoises.
25	25	Rainforest Diet	A rich variety of fruits, insects, and leaves formulated for tropical species such as tree frogs, toucans, and sloths.

AnimalDiet:

	AnimalID	DietID	StartDT	EndDT	FeedTM
1	1	1	NULL	NULL	NULL
2	1	2	NULL	NULL	NULL
3	2	1	NULL	NULL	NULL
4	3	2	NULL	NULL	NULL
5	4	3	NULL	NULL	NULL
6	5	1	NULL	NULL	NULL
7	6	1	NULL	NULL	NULL
8	7	14	NULL	NULL	NULL
9	8	14	NULL	NULL	NULL
10	9	3	NULL	NULL	NULL
11	10	3	NULL	NULL	NULL
12	11	2	NULL	NULL	NULL
13	12	2	NULL	NULL	NULL
14	13	16	NULL	NULL	NULL
15	14	16	NULL	NULL	NULL
16	15	3	NULL	NULL	NULL
17	15	14	NULL	NULL	NULL
18	16	3	NULL	NULL	NULL
19	16	14	NULL	NULL	NULL
20	17	1	NULL	NULL	NULL
21	18	1	NULL	NULL	NULL
22	19	9	NULL	NULL	NULL
23	20	9	NULL	NULL	NULL
24	21	14	NULL	NULL	NULL
25	22	14	NULL	NULL	NULL
26	23	11	NULL	NULL	NULL
27	24	11	NULL	NULL	NULL
28	25	2	NULL	NULL	NULL
29	26	2	NULL	NULL	NULL
30	27	9	NULL	NULL	NULL
31	28	9	NULL	NULL	NULL
32	29	1	NULL	NULL	NULL
33	30	1	NULL	NULL	NULL

Location:

	LocationID	LName	LDesc	TimeInserted	InsertBy
1	1	Jungle Land	Place for jungle dwellers	2025-11-06 10:04:50.823	ColtonsLaptop\ccsch
2	2	Aquarium	Place for water lovers	2025-11-06 10:09:30.507	ColtonsLaptop\ccsch

SQL 3

1) SQL to create a Customer table using appropriate data types that has the following columns: CustomerID, First Name, Last Name, Date of Birth, Address, City, State, Zip Code. CustomerID must be defined as an identity field that starts with 100.

```
--Creates a Customer table with eight different columns
Create Table Customer (
    CustomerID INT IDENTITY(100,1) NOT NULL,
    FirstName VARCHAR(50) NOT NULL,
    LastName VARCHAR(50) NOT NULL,
    DateOfBirth DATE,
    Address VARCHAR(100),
    City VARCHAR(50),
    State CHAR(2),
    ZipCode VARCHAR(10),
```

2) SQL to define the primary key in the CustomerID column. (SQL must not be generated)

```
CONSTRAINT PK_Customer PRIMARY KEY (CustomerID)
```

3) SQL to define an index using the primary key and a second index using Last Name and First Name. (SQL must not be generated)

```
Create NONCLUSTERED INDEX IX_Customer_LastName_FirstName
ON Customer (LastName, FirstName);
```

4) SQL to insert fifty rows of unique data into the Customer table.

```
INSERT INTO Customer (FirstName, LastName, DateOfBirth, Address, City, State, ZipCode)
VALUES
('John', 'Smith', '1980-01-15', '100 Main St', 'Springfield', 'IL', '62701'),
('Jane', 'Doe', '1981-02-10', '101 Oak Ave', 'Springfield', 'IL', '62702'),
('Michael', 'Johnson', '1982-03-12', '102 Pine Rd', 'Springfield', 'IL', '62703'),
('Emily', 'Davis', '1983-04-05', '103 Maple Dr', 'Springfield', 'IL', '62704'),
('William', 'Brown', '1984-05-25', '104 Cedar St', 'Springfield', 'IL', '62705'),
('Olivia', 'Wilson', '1985-06-13', '105 Birch Blvd', 'Springfield', 'IL', '62706'),
('James', 'Taylor', '1986-07-09', '106 Walnut Ln', 'Springfield', 'IL', '62707'),
('Sophia', 'Anderson', '1987-08-22', '107 Cherry Ct', 'Springfield', 'IL', '62708'),
('Benjamin', 'Thomas', '1988-09-14', '108 Poplar Way', 'Springfield', 'IL', '62709'),
('Ava', 'Jackson', '1989-10-30', '109 Aspen St', 'Springfield', 'IL', '62710'),
('Lucas', 'White', '1990-01-01', '110 Magnolia Ave', 'Springfield', 'IL', '62711'),
('Mia', 'Harris', '1991-02-14', '111 Elm St', 'Springfield', 'IL', '62712'),
('Henry', 'Martin', '1992-03-20', '112 Pinecrest Dr', 'Springfield', 'IL', '62713'),
('Amelia', 'Thompson', '1993-04-25', '113 Brookside Ln', 'Springfield', 'IL', '62714'),
('Alexander', 'Garcia', '1994-05-30', '114 Parkview Ct', 'Springfield', 'IL', '62715'),
('Charlotte', 'Martinez', '1995-06-12', '115 Hillcrest Rd', 'Springfield', 'IL', '62716'),
('Daniel', 'Robinson', '1996-07-08', '116 Riverbend St', 'Springfield', 'IL', '62717'),
('Harper', 'Clark', '1997-08-03', '117 Fairview Ave', 'Springfield', 'IL', '62718'),
('Matthew', 'Rodriguez', '1998-09-09', '118 Willow Way', 'Springfield', 'IL', '62719'),
('Evelyn', 'Lewis', '1999-10-10', '119 Sycamore Ct', 'Springfield', 'IL', '62720'),
('Joseph', 'Lee', '1980-11-15', '120 Ridge Rd', 'Springfield', 'IL', '62721'),
('Ella', 'Walker', '1981-12-01', '121 Highland Dr', 'Springfield', 'IL', '62722'),
('David', 'Hall', '1982-01-19', '122 Valley St', 'Springfield', 'IL', '62723'),
('Abigail', 'Allen', '1983-02-07', '123 Sunset Blvd', 'Springfield', 'IL', '62724'),
('Samuel', 'Young', '1984-03-26', '124 Lakeview Dr', 'Springfield', 'IL', '62725'),
('Isabella', 'Hernandez', '1985-04-11', '125 Forest Ct', 'Springfield', 'IL', '62726'),
('Andrew', 'King', '1986-05-04', '126 Meadow Ln', 'Springfield', 'IL', '62727'),
('Grace', 'Wright', '1987-06-15', '127 Orchard Rd', 'Springfield', 'IL', '62728'),
('Christopher', 'Lopez', '1988-07-20', '128 Grove St', 'Springfield', 'IL', '62729'),
('Chloe', 'Hill', '1989-08-29', '129 Birchwood Ave', 'Springfield', 'IL', '62730'),
('Joshua', 'Scott', '1990-09-10', '130 Spruce Ct', 'Springfield', 'IL', '62731'),
('Lily', 'Green', '1991-10-19', '131 Garden Rd', 'Springfield', 'IL', '62732'),
('Ryan', 'Adams', '1992-11-05', '132 Lake St', 'Springfield', 'IL', '62733'),
('Scarlett', 'Baker', '1993-12-09', '133 Ridgeview Dr', 'Springfield', 'IL', '62734'),
('Nathan', 'Nelson', '1994-01-22', '134 Maplewood Ct', 'Springfield', 'IL', '62735'),
('Victoria', 'Carter', '1995-02-18', '135 Pleasant St', 'Springfield', 'IL', '62736'),
('Anthony', 'Mitchell', '1996-03-10', '136 Cherrywood Ln', 'Springfield', 'IL', '62737'),
('Zoe', 'Perez', '1997-04-28', '137 Whispering Pines Dr', 'Springfield', 'IL', '62738'),
('Dylan', 'Roberts', '1998-05-16', '138 Woodland Ave', 'Springfield', 'IL', '62739'),
('Hannah', 'Turner', '1999-06-09', '139 Highland Ct', 'Springfield', 'IL', '62740'),
('Liam', 'Phillips', '1980-07-22', '140 North St', 'Springfield', 'IL', '62741'),
('Avery', 'Campbell', '1981-08-03', '141 South Ave', 'Springfield', 'IL', '62742'),
('Ethan', 'Parker', '1982-09-11', '142 East Blvd', 'Springfield', 'IL', '62743'),
('Sofia', 'Evans', '1983-10-05', '143 West Dr', 'Springfield', 'IL', '62744'),
('Logan', 'Edwards', '1984-11-19', '144 Meadowbrook Rd', 'Springfield', 'IL', '62745'),
('Aria', 'Collins', '1985-12-23', '145 Greenway Ct', 'Springfield', 'IL', '62746'),
('Jacob', 'Stewart', '1986-01-14', '146 Forestview Ln', 'Springfield', 'IL', '62747'),
('Layla', 'Sanchez', '1987-02-27', '147 River Rd', 'Springfield', 'IL', '62748'),
('Noah', 'Morris', '1988-03-15', '148 Valley View Dr', 'Springfield', 'IL', '62749'),
('Mila', 'Rogers', '1989-04-10', '149 Woodlawn Ave', 'Springfield', 'IL', '62750');
```

5) SQL to truncate and drop the Customer table.

```
Truncate Table Customer -- Removes data from the table while keeping the design
Drop Table Customer -- Deletes whole table
```

6) SQL to create a view of the Customer table using all of the columns in the table except Date of Birth.

```
Create View vw_Customer1 as
Select CustomerID, FirstName, LastName, Address, City, State, ZipCode
From Customer
```

```
Select * from vw_Customer1
```

	CustomerID	FirstName	LastName	Address	City	State	ZipCode
1	100	John	Smith	100 Main St	Springfield	IL	62701
2	101	Jane	Doe	101 Oak Ave	Springfield	IL	62702
3	102	Michael	Johnson	102 Pine Rd	Springfield	IL	62703
4	103	Emily	Davis	103 Maple Dr	Springfield	IL	62704
5	104	William	Brown	104 Cedar St	Springfield	IL	62705
6	105	Olivia	Wilson	105 Birch Blvd	Springfield	IL	62706
7	106	James	Taylor	106 Walnut Ln	Springfield	IL	62707
8	107	Sophia	Anderson	107 Cherry Ct	Springfield	IL	62708
9	108	Benjamin	Thomas	108 Poplar Way	Springfield	IL	62709
10	109	Ava	Jackson	109 Aspen St	Springfield	IL	62710
11	110	Lucas	White	110 Magnolia ...	Springfield	IL	62711
12	111	Mia	Harris	111 Elm St	Springfield	IL	62712
13	112	Henry	Martin	112 Pinecrest ...	Springfield	IL	62713
14	113	Amelia	Thomps...	113 Brooksid...	Springfield	IL	62714
15	114	Alexander	Garcia	114 Parkview ...	Springfield	IL	62715
16	115	Charlotte	Martinez	115 Hillcrest Rd	Springfield	IL	62716
17	116	Daniel	Robinson	116 Riverben...	Springfield	IL	62717
18	117	Harper	Clark	117 Fairview ...	Springfield	IL	62718
19	118	Matthew	Rodrigu...	118 Willow Way	Springfield	IL	62719
20	119	Evelyn	Lewis	119 Sycamor...	Springfield	IL	62720
21	120	Joseph	Lee	120 Ridge Rd	Springfield	IL	62721
22	121	Ella	Walker	121 Highland ...	Springfield	IL	62722
23	122	David	Hall	122 Valley St	Springfield	IL	62723
24	123	Abigail	Allen	123 Sunset Bl...	Springfield	IL	62724
25	124	Samuel	Young	124 Lakeview...	Springfield	IL	62725
26	125	Isabella	Hernand...	125 Forest Ct	Springfield	IL	62726
27	126	Andrew	King	126 Meadow ...	Springfield	IL	62727

7) SQL that uses the view to display all Customers in the Customer table sorted by First Name in descending order.

```
Select * from vw_Customer1
Order By FirstName desc
```

Results		Messages					
	CustomerID	FirstName	LastName	Address	City	State	ZipCode
1	137	Zoe	Perez	137 Whispering Pines Dr	Springfield	IL	62738
2	104	William	Brown	104 Cedar St	Springfield	IL	62705
3	135	Victoria	Carter	135 Pleasant St	Springfield	IL	62736
4	107	Sophia	Anderson	107 Cherry Ct	Springfield	IL	62708
5	143	Sofia	Evans	143 West Dr	Springfield	IL	62744
6	133	Scarlett	Baker	133 Ridgeview Dr	Springfield	IL	62734
7	124	Samuel	Young	124 Lakeview Dr	Springfield	IL	62725
8	132	Ryan	Adams	132 Lake St	Springfield	IL	62733
9	105	Olivia	Wilson	105 Birch Blvd	Springfield	IL	62706
10	148	Noah	Morris	148 Valley View Dr	Springfield	IL	62749
11	134	Nathan	Nelson	134 Maplewood Ct	Springfield	IL	62735
12	149	Mila	Rogers	149 Woodlawn Ave	Springfield	IL	62750
13	102	Michael	Johnson	102 Pine Rd	Springfield	IL	62703
14	111	Mia	Harris	111 Elm St	Springfield	IL	62712
15	118	Matthew	Rodriguez	118 Willow Way	Springfield	IL	62719
16	110	Lucas	White	110 Magnolia Ave	Springfield	IL	62711
17	144	Logan	Edwards	144 Meadowbrook Rd	Springfield	IL	62745
18	131	Lily	Green	131 Garden Rd	Springfield	IL	62732
19	140	Liam	Phillips	140 North St	Springfield	IL	62741
20	147	Layla	Sanchez	147 River Rd	Springfield	IL	62748
21	130	Joshua	Scott	130 Spruce Ct	Springfield	IL	62731
22	120	Joseph	Lee	120 Ridge Rd	Springfield	IL	62721
23	100	John	Smith	100 Main St	Springfield	IL	62701
24	101	Jane	Doe	101 Oak Ave	Springfield	IL	62702
25	106	James	Taylor	106 Walnut Ln	Springfield	IL	62707
26	146	Jacob	Stewart	146 Forestview Ln	Springfield	IL	62747
27	125	Isabella	Hernandez	125 Forest Ct	Springfield	IL	62726

8) One sql routine with at least 3 If statements and confirmation that they work.

```

Declare @Age int = 5
/ If @Age > 3
    Print 'Age GT 3. Age = ' + CAST(@Age as VARCHAR(10))
.
/ If @Age = 3
    Print 'Age = 3. Age = ' + CAST(@Age as VARCHAR(10))
.
/ If @Age < 3
    Print 'Age LT 3. Age = ' + CAST(@Age as VARCHAR(10))
.

```

Messages

Age GT 3. Age = 5

Completion time: 2025-11-11T13:34:36.7512562-05:00

9) One sql routine with If, Else-If and confirmation that they work.

```

Declare @Age int = 3
If @Age > 3
    Print 'Age GT 3. Age = ' + CAST(@Age as VARCHAR(10))
Else If @Age = 3
    Print 'Age = 3. Age = ' + CAST(@Age as VARCHAR(10))
Else
    Print 'Age LT 3. Age = ' + CAST(@Age as VARCHAR(10))

```

 Messages

Age = 3. Age = 3

Completion time: 2025-11-11T13:35:55.8693143-05:00

10) One Case statement with at least 3 When options and confirmation that they work.

```
Declare @Age int = 3

Print
Case
  When @Age > 3 Then 'Age GT 3. Age = ' + CAST(@Age as VARCHAR(10))
  When @Age = 3 Then 'Age = 3. Age = ' + CAST(@Age as VARCHAR(10))
  Else 'Age LT 3. Age = ' + CAST(@Age as VARCHAR(10))
END;
```

```
Declare @Age int = 0

Print
Case
  When @Age > 3 Then 'Age GT 3. Age = ' + CAST(@Age as VARCHAR(10))
  When @Age = 3 Then 'Age = 3. Age = ' + CAST(@Age as VARCHAR(10))
  Else 'Age LT 3. Age = ' + CAST(@Age as VARCHAR(10))
END;
```

 Messages

Age = 3. Age = 3

Completion time: 2025-11-11T13:37:43.7147005-05:00

 Messages

Age LT 3. Age = 0

Completion time: 2025-11-11T13:38:39.8110242-05:00

11) Alter table sql to add an integer column to an existing table that allows nulls.

```
Alter table Customer
Add Age int NULL; -- Allows the Age Column to have Null values
```

12) SQL update routine with error checking and error logging. Confirmation statement that generates logged error.

```
Declare @ErrorHold int

insert into Diet
(DietID, DName, DDesc)
Values
(1, 'Carnivore Diet', 'meat diet')
Set @ErrorHold = @@Error

Print '@@ErrorHold: ' + Cast(@ErrorHold as varchar(10))
```

Messages

Msg 2627, Level 14, State 1, Line 120
Violation of PRIMARY KEY constraint 'PK_Diet'. Cannot insert duplicate key in object 'dbo.Diet'. The statement has been terminated.
@@ErrorHold: 2627

Completion time: 2025-11-11T13:44:57.6090505-05:00

13) SQL statement that defines and uses variables.

```
Declare @Age int = 3

If @Age > 3
Print 'Age GT 3. Age = ' + Cast(@Age as VARCHAR(10))
Else If @Age = 3
Print 'Age = 3. Age = ' + Cast(@Age as VARCHAR(10))
Else
Print 'Age LT 3. Age = ' + Cast(@Age as VARCHAR(10))
```

14) Store procedure that accepts a variable and has error checking and logging as part of the execute stored procedure statement.

```
Declare @ErrorHold int -- Creates ErrorHold variable
Set @ErrorHold = @@Error -- Sets the variable to @@Error
Create Proc spu_InsertErrorAudit @ErrorHold int as --Creates stored procedure
Insert into ErrorAudit
(ErrorNo)
Values
(@ErrorHold)-- uses variable

Insert into Location
(LocationID, LName, LDesc)
Values
(1, 'Desert Land', 'Sand Lovers')-- Error Function
Set @ErrorHold = @@Error-- Sets ErrorHold to @@Error
If @ErrorHold != 0
exec spu_InsertErrorAudit @ErrorHold -- Executes stored procedure if an error occurs
```

15) SQL insert statement that inserts ten rows of data into a new table named Cars that includes at least 5 columns and the data varies for each record by using variables that are incremented and/or changed within the routine for each record.

```
Create Table Cars (
CarID INT IDENTITY(1,1) NOT NULL,
CName VARCHAR(50) NOT NULL,
LicencePlate Int NOT NULL,
Color VARCHAR(50) Null,
Brand VARCHAR(50) Not Null,

CONSTRAINT PK_Cars PRIMARY KEY (CarID)
);
insert into Cars
(CName, LicencePlate, Color, Brand)
Values
('Impala', 'EJK2345', 'Red', 'Chevrolet'),
('Civic', 'HND6789', 'Blue', 'Honda'),
('Corolla', 'TYT1011', 'Black', 'Toyota'),
('Mustang', 'FRD1213', 'Yellow', 'Ford'),
('A4', 'AUD1415', 'Grey', 'Audi'),
('Model 3', 'TSL1617', 'White', 'Tesla'),
('Golf', 'VWX1819', 'Green', 'Volkswagen'),
('Altima', 'NSN2021', 'Silver', 'Nissan'),
('G-Wagon', 'MCB2223', 'Matte Black', 'Mercedes'),
('Explorer', 'FRD2425', 'Metallic Blue', 'Ford')
```

16) Create a table named HikingLocations that has the appropriate structure to remove duplicates when loading the table. Demonstrate that it works. Include sql that defines the structure to remove duplicates in your submission. (this is the same as any other table create except you set the ignore_dup_key option to on).

```

Create Table HikingLocations (
    HikingLocationID INT IDENTITY(1,1) NOT NULL,
    HLName VARCHAR(100) NOT NULL,
    State VARCHAR(50) NOT NULL,

    CONSTRAINT PK_HikingLocations PRIMARY KEY Clustered (HikingLocationID)
    WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY = ON, ALLOW_ROW_LOCKS
) ON [PRIMARY]

--Could not get IGNORE_DUP_KEY = ON to work, so I researched to get this code.
CREATE UNIQUE NONCLUSTERED INDEX UQ_Location_State
ON HikingLocations (HLName, State)
WITH (IGNORE_DUP_KEY = ON);

Insert into HikingLocations
(HLName,State)
Values
('Teton National Park', 'Wyoming'),
('Yellowstone National Park', 'Wyoming'),
('Grand Canyon', 'Arizona')
Select * from HikingLocations

```

Messages

Duplicate key was ignored.

(0 rows affected)

Completion time: 2025-11-11T15:11:16.9906463-05:00

17) Define a table named HikingLocations2 that has 10 records and 3 of the records are duplicates. Create an SQL routine that identifies the duplicate records by key value. Create a second SQL routine that displays only the duplicate records.

hint:

```

SELECT a, b, COUNT(*) occurrences FROM t1 GROUP BY a, b HAVING COUNT(*) >
1;

```

```

SELECT HLName, State, COUNT(*) occurrences FROM HikingLocations2
GROUP BY HLName, State HAVING COUNT(*) > 1;

CREATE PROCEDURE DisplayOnlyDuplicateRecords AS
BEGIN
    WITH DuplicateKeys AS (
        -- Step 1: Identify all unique keys that appear more than once
        SELECT HLName, State
        FROM HikingLocations2
        GROUP BY HLName, State
        HAVING COUNT(*) > 1
    )
    -- Step 2: Join the original table back to the list of duplicate keys
    -- to retrieve all columns for every row that matches a duplicate key.
    SELECT H.HikingLocationID, H.HLName, H.State
    FROM HikingLocations2 AS H
    INNER JOIN
        DuplicateKeys AS DK ON
        H.HLName = DK.HLName AND
        H.State = DK.State
    ORDER BY
        H.HLName, H.HikingLocationID;
END
GO
Exec DisplayOnlyDuplicateRecords

```

Results		Messages	
	HLName	State	occurrences
1	Grand Canyon	Arizona	2
2	Teton National Park	Wyoming	2
3	Yellowstone National Park	Wyoming	2

Results		Messages	
	HikingLocationID	HLName	State
1	3	Grand Canyon	Arizona
2	10	Grand Canyon	Arizona
3	1	Teton National Park	Wyoming
4	8	Teton National Park	Wyoming
5	2	Yellowstone National Park	Wyoming
6	9	Yellowstone National Park	Wyoming

18) Create an insert routine for HikingLocations2 that uses variables and performs error checking and error logging.

```

Create Proc spu_InsertErrorAudit @ErrorHold int as
Insert into ErrorAudit
    (ErrorNo)
Values
    (@ErrorHold)
Set @ErrorHold = @@Error
If @ErrorHold != 0
    exec spu_InsertErrorAudit @ErrorHold

```

Overview:

This SQL assignment covers a lot of important SQL scripts that can be important in a database. The script that we used today, which I think is the most important, is the Error Auditing section. Being able to set up a procedure like this that can create error logs can be a huge benefit for a database's security. I was confused on question 17, so I used the internet to help me come up with a script. Identifying versus displaying confused me for some reason. I enjoyed this assignment and think that it is a great, thought-out assignment.